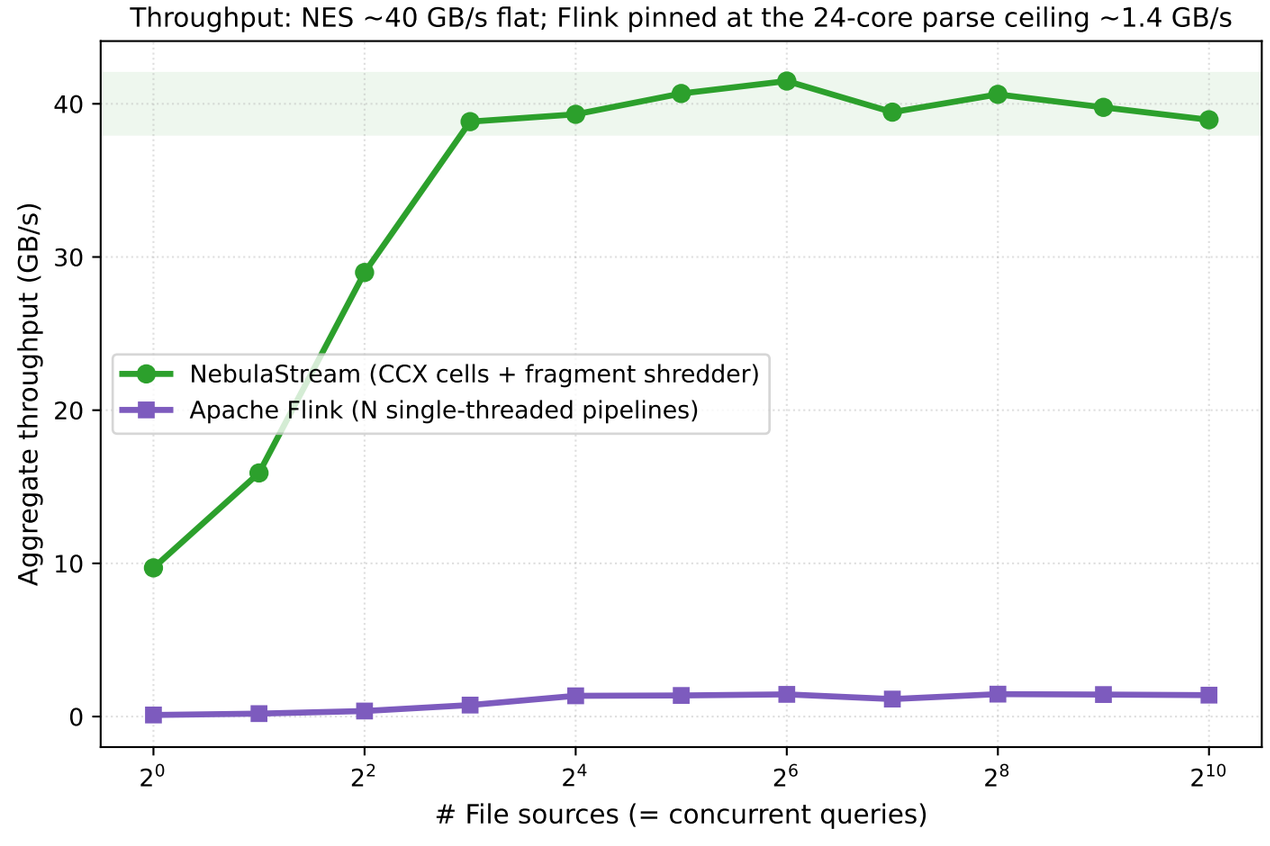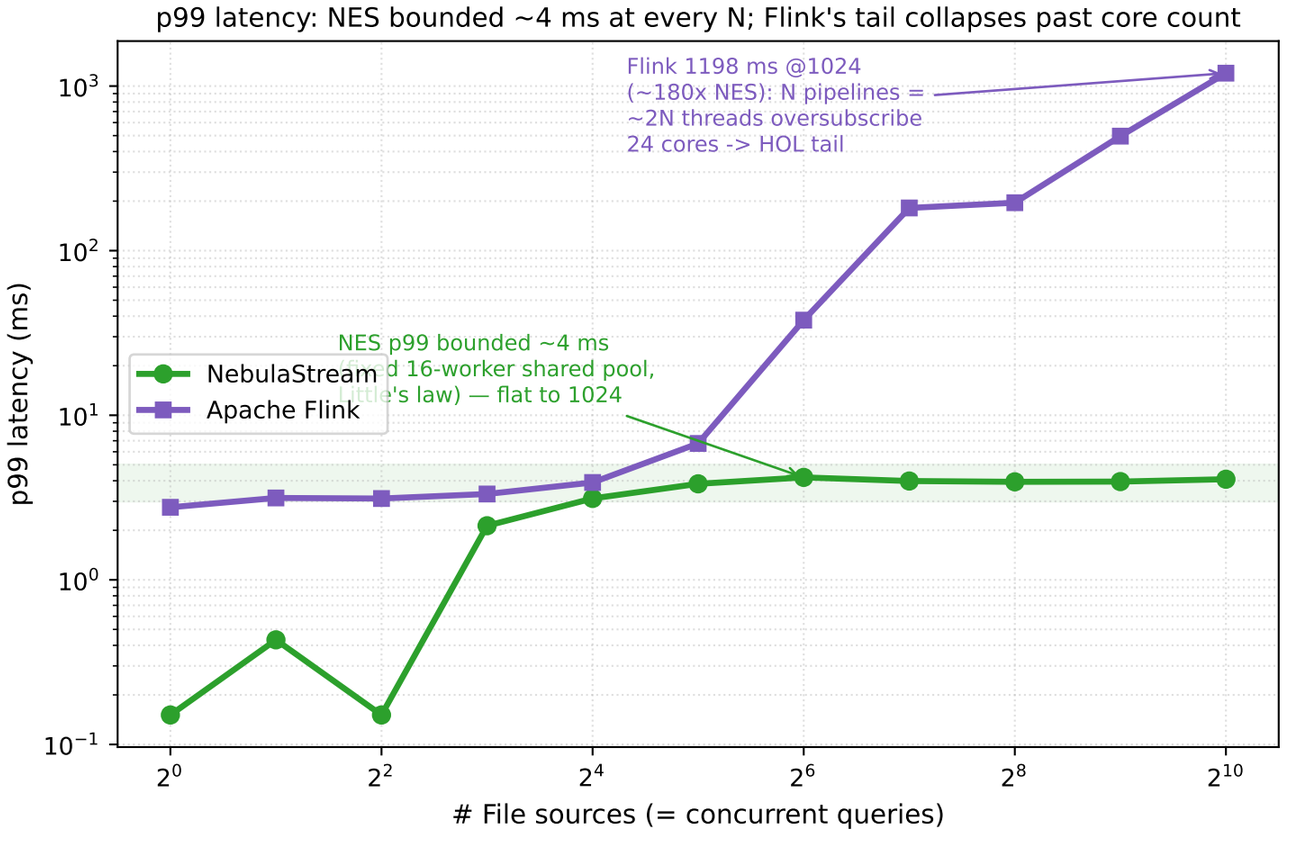File-source scaling, 1 -> 1024 sources: NebulaStream (CCX cells + fragment shredder) vs Apache Flink
bid passthrough, warm page cache — e2e ingest->sink latency, same per-buffer methodology on both engines